Random generalized Lotka-Volterra model on sparse graphs: Gaussian expansion of dynamic cavity method

Mattia Tarabolo,[1,*] A. R. Batista-Tomás,[2] Roberto Mulet,[2] and Luca Dall'Asta[1,3,4]

[1] *Politecnico di Torino, Torino, Italy*
[2] *University of Havana, La Habana, Cuba*
[3] *Collegio Carlo Alberto, Torino, Italy*
[4] *Italian Institute for Genomic Medicine (IIGM), Candiolo, Italy*

(Dated: November 3, 2025)

Abstract

An article usually includes an abstract, a concise summary of the work covered at length in the main body of the article.

I. INTRODUCTION

Intro on the subject.

II. METHOD

A. Random generalized Lotka-Volterra model on sparse graphs

We consider a generalized Lotka-Volterra (gLV) model describing the dynamics of an interacting community of N species, labeled by $i = 1, \ldots, N$. The species interacts on an undirected random graph G with symmetric adjacency matrix $A = \{a_{ij}\}$, meaning that species i and j are interacting if $a_{ij} = a_{ji} = 1$, otherwise the matrix elements are zero. The time-dependent number density of individuals of species i is denoted by $x_i(t)$, and evolves in time according to

$$\frac{d}{dt}x_i(t) = x_i(t)\left[1 - x_i(t) + \sum_{j\neq i} a_{ij}J_{ij}x_j(t)\right] + \eta_i(t), \tag{1}$$

where we included, for the sake of generality, a Gaussian noise $\eta_i(t)$ vanishing mean and variance $\langle \eta_i(t)\eta_{i'}(t')\rangle = 2Tg(x_i(t))\delta_{ii'}\delta(t-t')$. We will consider in what follows three choices for the function $g(x)$: (i) $g(x) = 0$ for deterministic gLV model, (ii) $g(x) = x$ for demographic noise, and (iii) $g(x) = x^2$ for environmental noise.

The couplings J_{ij} represent the (per capita) effect of species on one another. A negative coefficient J_{ij} indicates a competitive effect of species j on species i.

We consider here Gaussian J_{ij} with average $\overline{J_{ij}} = \overline{J_{ji}} = m/K$, variance $\mathrm{Var}(J_{ij}) = \mathrm{Var}(J_{ji}) = \sigma^2/K$ and covariance $\mathrm{Corr}(J_{ij}, J_{ji}) = \gamma$. The overbar $\overline{\cdots}$ represents the average over the random couplings ensemble. The scaling of the moments with the average degree of the graph K is needed to produce a well-defined fully connected limit $K \to \infty$. The parameter $-1 \leq \gamma \leq 1$, being related to the correlation between the interaction coefficients J_{ij} and J_{ji}, measures the fraction of predator-prey pairs. Such a predator-prey pair consists of two species i and j for which J_{ij} and J_{ji} have opposite signs. If for example J_{ij} is positive and J_{ji} is negative, the presence of species i has a detrimental effect on species j, while the presence of species j is beneficial for species i. For any pair $i < j$ we can write

$$J_{ij} = \frac{m}{K} + \frac{\sigma}{\sqrt{K}}z_{ij}, \qquad J_{ji} = \frac{m}{K} + \frac{\sigma}{\sqrt{K}}z_{ji}, \tag{2}$$

where z_{ij} and z_{ji} are Gaussian random variables with $\overline{z_{ij}} = 0$, $\overline{z_{ij}^2} = 1$, $\overline{z_{ij}z_{ji}} = \gamma$.

For $\gamma = 1$ we have $z_{ij} = z_{ji}$ with probability one, for $\gamma = 0$ z_{ij} and z_{ji} are uncorrelated, and for $\gamma = -1$ we have $z_{ij} = z_{ji}$ with probability one. The latter, $\gamma = -1$, corresponds to having all the interactions of the predator-prey type. Increasing the value of γ this fraction decrease, until the opposite case , $\gamma = 1$ of no predator-prey type interactions. The other interactions can be of a mutualistic type (J_{ij} and J_{ji} both positive) or strictly competitive (J_{ij} and J_{ji} both negative).

B. Gaussian approximation of the dynamic cavity equations on a Random Regular Graph (RRG)

We focus here on the case of a Random Regular Graph (RRG) G, where edges are randomly extracted with the constraint that each node has exactly K neighbors.

We use the Gaussian Expansion Cavity method [] to derive a Langevin equation for a representative species. This approach begins by constructing the Martin-Siggia-Rose-Janssen-De Dominicis (MSRJD) dynamic partition function through path integrals of the trajectories [? ? ?]

$$Z = \left\langle \int \mathcal{D}\vec{x} \prod_i p_i\left(x_i\left(0\right)\right) \delta\left(\dot{x_i}(t) - x_i(t)\left[1 - x_i(t) + \sum_{j\neq i} a_{ij} J_{ij} x_j(t) + h_i(t)\right] - \eta_i(t)\right)\right\rangle_{\vec{\eta}}, \tag{3}$$

where $\int \mathcal{D}\vec{x} = \prod_i \int \mathcal{D}x_i$ is a functional integral over the trajectories of the system, and $h_i(t)$ is an external field, needed to compute response functions. Following the steps of [?] we put forward a dynamic cavity Ansatz and expand the cavity marginals for small couplings. This results in a set of effective de-coupled Langevin equations on the cavity graph. For any edge (i,j) on the graph G the effective cavity population size $x_{i\setminus j}(t)$ evolves under the following stochastic process

$$\begin{aligned}\frac{d}{dt}x_{i\setminus j}(t) = x_{i\setminus j}(t)\Bigg[1 - x_{i\setminus j}(t) + \sum_{k\in\partial i\setminus j} J_{ik}\mu_{k\setminus i}(t) + \sum_{k\in\partial i\setminus j}\int_0^t dt' J_{ik}J_{ki}R_{k\setminus i}(t,t')x_{i\setminus j}(t') \\ + h_{i\setminus j}(t) + \zeta_{i\setminus j}(t)\Bigg] + \xi_{i\setminus j}(t)\end{aligned} \tag{4}$$

where $h_{i\setminus j}(t)$ is an external field, needed to compute response functions, and $\zeta_{i\setminus j}$ and $\xi_{i\setminus j}$ are Gaussian noises with vanishing mean and correlations

$$\left\langle\zeta_{i\setminus j}(t)\zeta_{i\setminus j}(t')\right\rangle_{i\setminus j} = \sum_{k\in\partial i\setminus j} J_{ik}^2 C_{k\setminus i}(t,t'), \qquad \left\langle\xi_{i\setminus j}(t)\xi_{i\setminus j}(t')\right\rangle_{i\setminus j} = g(x_{i\setminus j}(t))\delta(t-t'). \tag{5}$$

The terms $\mu_{k\setminus i}(t)$, $R_{k\setminus i}(t,t')$, and $C_{k\setminus i}(t,t')$ are respectively the cavity mean abundance, the cavity response function, and the cavity (connected) two-point correlation function. They are obtained self-consistently from the effective dynamics of the neighbors

$$\mu_{k\setminus i}(t) = \left\langle x_{k\setminus i}(t)\right\rangle_{k\setminus i}, \quad R_{k\setminus i}(t,t') = \frac{\delta}{\delta h_{k\setminus i}(t')}\left\langle x_{k\setminus i}(t)\right\rangle_{k\setminus i}, \quad C(t,t') = \left\langle x_{k\setminus i}(t)x_{k\setminus i}(t')\right\rangle_{k\setminus i}, \tag{6}$$

where $\langle\dots\rangle_{k\setminus i}$ describes the average over the distribution of the effective process Eq. (4) for the edge (k,i). The effective process is therefore de-coupled, and the interactions are captured by the self-consistent external fields $\mu_{k\setminus i}(t)$, $R_{k\setminus i}(t,t')$, and $C_{k\setminus i}(t,t')$. A detailed derivation can be found in Appendix A.

Once the cavity averages have been determined through self-consistency, one can obtain the effective de-coupled Langevin equations for the species population

$$\begin{aligned}\frac{d}{dt}x_i(t) = x_i(t)\Bigg[1 - x_i(t) + \sum_{k\in\partial i} J_{ik}\mu_{k\setminus i}(t) + \sum_{k\in\partial i}\int_0^t dt' J_{ik}J_{ki}R_{k\setminus i}(t,t')x_i(t') \\ + h_i(t) + \zeta_i(t)\Bigg] + \xi_i(t),\end{aligned} \tag{7}$$

with Gaussian noises with vanishing mean and correlations

$$\langle\zeta_i(t)\zeta_i(t')\rangle_i = \sum_{k\in\partial i} J_{ik}^2 C_{k\setminus i}(t,t'), \qquad \langle\xi_i(t)\xi_i(t')\rangle_i = g(x_i(t))\delta(t-t'). \tag{8}$$

The average $\langle\dots\rangle_i$ is performed over the effective de-coupled process Eq. (7).

C. Naive method: Gaussian expansion

The effective processes can be averaged over the disorder of both the RRG and the random couplings. Due to homogeneity of the system, this yields an effective process for a representative cavity species abundance $x_c(t)$

$$\frac{d}{dt}x_c(t) = x_c(t)\Bigg[1 - x_c(t) + \frac{K-1}{K}m\mu_c(t) + \frac{K-1}{K}\left(\frac{m^2}{K}+\gamma\sigma^2\right)\int_0^t dt' R_c(t,t')x_c(t') + \xi_c(t) + h_c(t)\Bigg], \tag{9}$$

where the Gaussian colored noise has mean and covariance

$$\langle\xi_c(t)\rangle_c = 0, \qquad \langle\xi_c(t)\xi_c(t')\rangle_c = 2T\delta(t-t') + \frac{K-1}{K}\left(\frac{m^2}{K}+\sigma^2\right)C_c(t,t').$$

The representative species abundance in the original graph evolves under the following effective process

$$\frac{d}{dt}x(t) = x(t)\left[1 - x(t) + m\mu_c(t) + \left(\frac{m^2}{K}+\gamma\sigma^2\right)\int_0^t dt' R_c(t,t')x(t') + \xi(t) + h(t)\right], \tag{10}$$

where the Gaussian colored noise has mean and covariance

$$\langle\xi(t)\rangle = 0, \qquad \langle\xi(t)\xi(t')\rangle = 2T\delta(t-t') + \left(\frac{m^2}{K}+\sigma^2\right)C_c(t,t'). \tag{11}$$

It seems that the averaging over disorder was done in a wrong way. We show however that this corresponds to a Gaussian approximation of the disorder averaged dynamical partition function. The calculations can be found in Appendix B. The problem is that this approximation corresponds to a large connectivity expansion, already done by [?].

1. *Fixed point analysis*

In the limit $T \to 0$ we can assume that the system reaches a steady state which does not depends on the initial condition. We further assume that the dynamics reaches a unique fixed-point $\lim_{t\to\infty} x_c(t) = x_c^*$, $\lim_{t\to\infty} x(t) = x^*$, and therefore $\lim_{t\to\infty}\xi_c(t) = \xi_c^*$, and $\lim_{t\to\infty}\xi(t) = \xi^*$. The fixed point values x_c^*, x^*, ξ_c^*, and ξ^* are all random variables. It follows that the mean cavity abundance and the cavity correlation become become time-independent random variables $\lim_{t\to\infty}\mu_c(t) = \mu_c^* = \langle x_c^*\rangle_c^*$, and $\lim_{t,t'\to\infty} C_c(t,t') = q_c^* = \langle x_c^* x_c^*\rangle_c^*$. Within this assumptions the cavity response function is time-translational-invariant, i.e., $R_c(t,t')$ depends only on the time difference $\tau = t - t'$. Due to causality $R_c(\tau) = 0$ for $\tau \leq 0$, and we define $\chi_c^* = \int_0^\infty d\tau\, C_c^*(\tau)$.

The fixed point satisfies the self-consistent equation

$$x_c^*\left[1 - x_c^* + \frac{K-1}{K}m\mu_c^* + \frac{K-1}{K}\left(\frac{m^2}{K}+\gamma\sigma^2\right)\chi_c^* x_c^* + \xi_c^* + h_c\right] = 0, \tag{12}$$

where h_c is a constant external field ($h_c(t) = h_c$) needed to obtain the integrated cavity response. The fixed point cavity noise is a Gaussian random variable with mean zero and variance $\Sigma_c^2 q_c^* = (K-1)/K(m^2/K+\sigma^2)q_c^*$. We can therefore write is as $\xi_c^* = -\Sigma_c\sqrt{q_c^*}s$, where s is a standard Gaussian random variable with zero mean and unit variance. Eq. (12) admits non-vanishing solution

$$x_c^*(s) = \frac{\Sigma_c\sqrt{q_c^*}(\Delta_c - s) + h_c}{1 - \frac{K-1}{K}\left(\frac{m^2}{K}+\gamma\sigma^2\right)\chi_c^*}\Theta\left(\frac{\Sigma_c\sqrt{q_c^*}(\Delta_c - s) + h_c}{1 - \frac{K-1}{K}\left(\frac{m^2}{K}+\gamma\sigma^2\right)\chi_c^*}\right), \tag{13}$$

where $\Theta(x)$ is the Heaviside step function, which ensures positivity of the fixed-point cavity abundance. For simplicity of notation we have used the abbreviation $\Delta_c = (1 + \frac{K-1}{K} m\mu_c^*)/(\Sigma_c\sqrt{q_c^*})$. In a similar way, one finds the fixed-point representative abundance

$$x^*(\xi^*) = \frac{1 + m\mu_c^* + \xi^*}{1 - \left(\frac{m^2}{K} + \gamma\sigma^2\right)\chi_c^*}\Theta\left(\frac{1 + m\mu_c^* + \xi^*}{1 - \left(\frac{m^2}{K} + \gamma\sigma^2\right)\chi_c^*}\right). \tag{14}$$

We perform the average over the ensemble of fixed points following the derivation of [1**?** , 2], obtaining the integral equations (see Appendix C for a complete derivation)

$$\mu_c^* = \frac{\Sigma_c\sqrt{q_c^*}}{1 - \frac{K-1}{K}\left(\frac{m^2}{K} + \gamma\sigma^2\right)\chi_c^*}\int_{-\infty}^{\Delta_c} Ds\ (\Delta_c - s) \tag{15a}$$

$$1 = \frac{\Sigma_c^2}{\left[1 - \frac{K-1}{K}\left(\frac{m^2}{K} + \gamma\sigma^2\right)\chi_c^*\right]^2}\int_{-\infty}^{\Delta_c} Ds\ (\Delta_c - s)^2 \tag{15b}$$

$$\chi_c^* = \frac{1}{1 - \frac{K-1}{K}\left(\frac{m^2}{K} + \gamma\sigma^2\right)\chi_c^*}\int_{-\infty}^{\Delta_c} Ds, \tag{15c}$$

where $Ds = e^{-s^2/2}/\sqrt{2\pi}$.

We recall that the integral equations are valid only in the stable fixed point phase, and require $1 - \frac{K-1}{K}\left(\frac{m^2}{K} + \gamma\sigma^2\right)\chi_c^* > 0$. Moreover, being μ_c^* the mean species abundance in the cavity process, it must be non-negative. The integral equations can be solved by Newton-Raphson methods, and deliver the cavity quantities μ_c^*, q_c^* and χ_c^* as a function of the model parameters σ, m and γ. Once those quantities have been computed, they can be inserted into Eq. (14) to obtain the fixed point distribution of the representative species abundance x^*, which has a delta-peak at zero, representing the fraction of species that go extinct, plus a clipped Gaussian, representing the distribution of surviving species,

$$P(x^*) = (1 - \phi)\delta(x^*) + p_{\text{surv}}(x^*). \tag{16}$$

The distribution of surviving species is a clipped Gaussian

$$p_{\text{surv}}(x^*) = \frac{1}{\sqrt{2\pi\Sigma_{\text{surv}}^2 q_c^*}} e^{-\frac{(x^* - x_S)^2}{2\Sigma_{\text{surv}}^2 q_c^*}}\Theta(x^*), \tag{17}$$

with

$$x_S = \frac{1 + m\mu_c^*}{1 - (m^2/K + \gamma\sigma^2)\chi_c^*}, \qquad \Sigma_{\text{surv}} = \frac{m^2/K + \sigma^2}{1 - (m^2/K + \gamma\sigma^2)\chi_c^*}. \tag{18}$$

The weight of the delta-peak at $x^* = 0$ is the fraction of species that go extinct $1 - \phi$. Consequently the fraction of surviving species is $\phi = \int_0^\infty dx^*\, p_{\text{surv}}(x^*) = \left(1 + \text{erf}(\Delta/\sqrt{2})\right)$ with $\Delta = (1 + m\mu_c^*) / \left((m^2/K + \sigma^2)\sqrt{q_c^*}\right)$.

2. Stability analysis

The fixed points we have studied in Section II C 1 are defined only in the stable phase of the Lotka-Volterra system, where the representative abundance converges to a unique stable fixed point. We study the stability of those fixed points through a linear stability analysis, in order to determine the boundaries of such a stable phase in the phase diagram. A detailed derivation can be found in Appendix D, where we adapted the derivation of [1**?**]. Once the dynamics Eq. (4) reaches the non-vanishing fixed point x_c^*, a small fluctuating perturbation $\varepsilon\zeta(t)$ is switched on, resulting in a fluctuation $y(t)$ about x_c^* which evolves under the stochastic process

$$\frac{d}{dt}y(t) = x_c^*\left[-y(t) + \frac{K-1}{K}\left(\frac{m^2}{K} + \gamma\sigma^2\right)\int_0^t dt'\, R_c^*(t,t')y(t') + \nu(t) + \zeta(t)\right], \tag{19}$$

where $\nu(t)$ is the deviation about the noise ξ_c^*. The perturbing noise $\zeta(t)$ is Gaussian with zero mean and unit variance. We can safely assume that $\nu(t)$ and $\zeta(t)$ are independent from x_c^* and ξ_c^*. Then one finds from self-consistency $\langle\nu(t)\nu(t')\rangle = \frac{K-1}{K}(\frac{m^2}{K}+\sigma^2)\langle y(t)y(t')\rangle$. Passing to the Fourier space in Eq. (19)

$$\tilde{y}(\omega) = \frac{\tilde{\nu}(\omega)+\tilde{\zeta}(\omega)}{\frac{\mathrm{i}\omega}{x_c^*} + \left(1 - \frac{K-1}{K}\left(\frac{m^2}{K}+\gamma\sigma^2\right)\tilde{R}_c^*(\omega)\right)}. \tag{20}$$

Focusing on the long-time behaviour of the correlation of fluctuations we find

$$\langle|\tilde{y}(0)|^2\rangle = \frac{\phi_c}{\left[1-\frac{K-1}{K}\left(\frac{m^2}{K}+\gamma\sigma^2\right)\chi_c^*\right]^2 - \frac{K-1}{K}\left(\frac{m^2}{K}+\sigma^2\right)\phi_c}, \tag{21}$$

where $\phi_c = \int_{-\infty}^{\Delta_c} Ds$ is the fraction of surviving species in the cavity process. The correlation diverges when $\frac{K-1}{K}\left(\frac{m^2}{K}+\sigma^2\right)\phi_c = \left[1-\frac{K-1}{K}\left(\frac{m^2}{K}+\gamma\sigma^2\right)\chi_c^*\right]^2$, which leads to $\Delta_c = 0$ and $\phi_c = 1/2$ from Eq. (15b). From this we find that the system is in the stable phase for

$$\frac{\left(2\frac{m^2}{K}+(1+\gamma)\sigma^2\right)^2}{\frac{m^2}{K}+\sigma^2} < 2\frac{K}{K-1}. \tag{22}$$

The result is consistent with the Fully Connected (FC) case, which corresponds to taking the limit of infinite connectivity $K\to\infty$. In that case the stable-fixed-point phase is independent of the value of m, and is determined by $\sigma^{\mathrm{FC}} < \sigma_c^{\mathrm{FC}}$, where $\left(\sigma_c^{\mathrm{FC}}\right)^2 = 2/(1+\gamma)^2$.

We note that sparseness reduce the stability of the system when only predator-prey interactions are present, i.e. when $\gamma=-1$. In that limit the system is stable when $(\sigma^{\mathrm{pp}})^2 > (\sigma_c^{\mathrm{pp}})^2 = \frac{m^2}{K}\left(2\frac{K-1}{K^2}m^2-1\right)$, which tends to zero in the FC limit, making the system stable in the whole phase space.

D. Proper derivation: population dynamics

In the limit $T\to 0$ we can assume that the system reaches a steady state which does not depends on the initial condition. We further assume that the dynamics reaches a unique fixed-point $\lim_{t\to\infty} x_{i\setminus j}(t) = x_{i\setminus j}^*$, and therefore $\lim_{t\to\infty}\xi_{i\setminus j}(t) = \xi_{i\setminus j}^*$. The fixed point values $x_{i\setminus j}^*$, $\xi_{i\setminus j}^*$, are all random variables. It follows that the mean cavity abundance and the cavity correlation become become time-independent random variables $\lim_{t\to\infty}\mu_{i\setminus j}(t) = \mu_{i\setminus j}^* = \langle x_{i\setminus j}^*\rangle_{i\setminus j}^*$, and $\lim_{t,t'\to\infty} C_{i\setminus j}(t,t') = q_{i\setminus j}^* = \langle(x_{i\setminus j}^*)^2\rangle_{i\setminus j}^*$. Within this assumptions the cavity response function is time-translational-invariant, i.e., $R_{i\setminus j}(t,t')$ depends only on the time difference $\tau = t-t'$. Due to causality $R_c(i\setminus j) = 0$ for $\tau\le 0$, and we define $\chi_{i\setminus j}^* = \int_0^\infty d\tau\, R_{i\setminus j}^*(\tau)$.

The fixed point satisfies the self-consistent equation

$$x_{i\setminus j}^*\left[1 - x_{i\setminus j}^* + \sum_{k\in\partial i\setminus j} J_{ik}\mu_{k\setminus i}^* + \sum_{k\in\partial i\setminus j} J_{ik}J_{ki}\chi_{k\setminus i}^* x_{i\setminus j}^* + \xi_{i\setminus j}^* + h_{i\setminus j}\right] = 0, \tag{23}$$

where $h_{i\setminus j}$ is a constant external field needed to obtain the integrated cavity response, and the noise is Gaussian with zero mean $\langle\xi_{i\setminus j}^*\rangle$ and variance $\langle(\xi_{i\setminus j})^*\rangle = \sum_{k\in\partial i\setminus j} J_{ik}^2 q_{k\setminus i}^* = \Sigma_{i\setminus j}^2$. We can therefore write $\xi_{i\setminus j}^* = -\Sigma_{i\setminus j}s$, where s is a standard Gaussian variable with zero mean and unit variance. Eq. (23) admits non-vanishing solution

$$x_{i\setminus j}^*(s) = \frac{\Sigma_{i\setminus j}(\Delta_{i\setminus j}-s)+h_{i\setminus j}}{\Gamma_{i\setminus j}}\Theta\left(\frac{\Sigma_{i\setminus j}(\Delta_{i\setminus j}-s)+h_{i\setminus j}}{\Gamma_{i\setminus j}}\right), \tag{24}$$

where $\Theta(x)$ is the Heaviside step function, which ensures positivity of the fixed-point cavity abundance. We have defined $\Delta_{i\setminus j} = (1+\sum_{k\in\partial i\setminus j} J_{ik}\mu_{k\setminus i}^*)/\Sigma_{i\setminus j}$, $\Gamma_{i\setminus j} = 1 - \sum_{k\in\partial i\setminus j} J_{ik}J_{ki}\chi_{k\setminus i}^*$ and we have assumed that $\sum_{k\in\partial i\setminus j} J_{ik}J_{ki}\chi_{k\setminus i}^* < 1$.

By averaging over the Gaussian distribution of s we obtain a set of self-consistent equations for the parameters

$$\mu^*_{i\setminus j} = \frac{\Sigma_{i\setminus j}}{\Gamma_{i\setminus j}} \left[\Theta(\Gamma_{i\setminus j}) \int_{-\infty}^{\Delta_{i\setminus j}} Ds \, (\Delta_{i\setminus j} - s) + \Theta(-\Gamma_{i\setminus j}) \int_{\Delta_{i\setminus j}}^{\infty} Ds \, (\Delta_{i\setminus j} - s) \right] \tag{25a}$$

$$q^*_{i\setminus j} = \frac{\Sigma^2_{i\setminus j}}{\Gamma^2_{i\setminus j}} \left[\Theta(\Gamma_{i\setminus j}) \int_{-\infty}^{\Delta_{i\setminus j}} Ds \, (\Delta_{i\setminus j} - s)^2 + \Theta(-\Gamma_{i\setminus j}) \int_{\Delta_{i\setminus j}}^{\infty} Ds \, (\Delta_{i\setminus j} - s)^2 \right] \tag{25b}$$

$$\chi^*_{i\setminus j} = \frac{1}{\Gamma_{i\setminus j}} \left[\Theta(\Gamma_{i\setminus j}) \int_{-\infty}^{\Delta_{i\setminus j}} Ds + \Theta(-\Gamma_{i\setminus j}) \int_{\Delta_{i\setminus j}}^{\infty} Ds \right]. \tag{25c}$$

We observe that the integrals can be written in terms of the error function $\mathrm{erf}(x) = \int_0^x ds \, e^{-s^2} 2/\sqrt{\pi}$. Defining $\varphi(x) = e^{-x^2/2}/\sqrt{2\pi}$ the normal distribution function and $\Phi(x) = [1 + \mathrm{erf}(x/\sqrt{2})]/2$ its cumulative distribution function we obtain

$$\mu^*_{i\setminus j} = \frac{\Sigma_{i\setminus j}}{\Gamma_{i\setminus j}} \left[\Theta(\Gamma_{i\setminus j}) \left(\Delta_{i\setminus j} \Phi(\Delta_{i\setminus j}) + \varphi(\Delta_{i\setminus j}) \right) + \Theta(-\Gamma_{i\setminus j}) \left(\Delta_{i\setminus j} \Phi(-\Delta_{i\setminus j}) - \varphi(-\Delta_{i\setminus j}) \right) \right] \tag{26a}$$

$$\begin{aligned} q^*_{i\setminus j} = \frac{\Sigma^2_{i\setminus j}}{\Gamma^2_{i\setminus j}} & \Big[\Theta(\Gamma_{i\setminus j}) \left(\left(1 + \Delta^2_{i\setminus j} \right) \Phi(\Delta_{i\setminus j}) + \Delta_{i\setminus j} \varphi(\Delta_{i\setminus j}) \right) \\ & + \Theta(\Gamma_{i\setminus j}) \left(\left(1 + \Delta^2_{i\setminus j} \right) \Phi(-\Delta_{i\setminus j}) - \Delta_{i\setminus j} \varphi(-\Delta_{i\setminus j}) \right) \Big] \end{aligned} \tag{26b}$$

$$\chi^*_{i\setminus j} = \frac{1}{\Gamma_{i\setminus j}} \left[\Theta(\Gamma_{i\setminus j}) \Phi(\Delta_{i\setminus j}) + \Theta(-\Gamma_{i\setminus j}) \Phi(-\Delta_{i\setminus j}) \right]. \tag{26c}$$

We can solve these equations by means of a population dynamics algorithm, here is their explicit version, together with the definitions of $\Delta_{i\setminus j}$ and $\Sigma_{i\setminus j}$

$$\Delta_{i\setminus j} = \frac{1 + \sum_{k \in \partial i \setminus j} J_{ik} \mu^*_{k\setminus i}}{\sqrt{\sum_{k \in \partial i \setminus j} J^2_{ik} q^*_{k\setminus i}}} \tag{27a}$$

$$\Sigma^2_{i\setminus j} = \sum_{k \in \partial i \setminus j} J^2_{ik} q^*_{k\setminus i} \tag{27b}$$

$$\Gamma_{i\setminus j} = 1 - \sum_{k \in \partial i \setminus j} J_{ik} J_{ki} \chi^*_{k\setminus i}. \tag{27c}$$

E. Proper derivation: just an idea...

From the effective process of the cavity population density Eq. (4) one can obtain dynamical self-consistent equations for the averages $\mu_{i\setminus j}$, $C_{i\setminus j}$, and $R_{i\setminus j}$.

$$\begin{aligned} \frac{d}{dt} \mu_{i\setminus j}(t) = \langle \dot{x}_{i\setminus j}(t) \rangle_{i\setminus j} &= \mu_{i\setminus j}(t) - C_{i\setminus j}(t,t) + \sum_{k \in \partial i \setminus j} J_{ik} \mu_{k\setminus i}(t) \mu_{i\setminus j}(t) \\ &+ \sum_{k \in \partial i \setminus j} \int_0^t dt' \, J_{ik} J_{ki} R_{k\setminus i}(t,t') C_{i\setminus j}(t,t') + \langle x_{i\setminus j}(t) \xi_{i\setminus j}(t) \rangle_{i\setminus j} \end{aligned} \tag{28a}$$

$$\begin{aligned} &= \mu_{i\setminus j}(t) - C_{i\setminus j}(t,t) + \sum_{k \in \partial i \setminus j} J_{ik} \mu_{k\setminus i}(t) \mu_{i\setminus j}(t) \\ &+ \sum_{k \in \partial i \setminus j} \int_0^t dt' \left[J_{ik} J_{ki} R_{k\setminus i}(t,t') C_{i\setminus j}(t,t') + J^2_{ik} C_{k\setminus i}(t,t') R_{i\setminus j}(t,t') \right], \end{aligned} \tag{28b}$$

where we used

$$\langle x_{i\setminus j}(t)\xi_{i\setminus j}(t')\rangle_{i\setminus j} = \frac{\delta}{\delta \mathrm{i}\hat{x}_{i\setminus j}(t')}\langle x_{i\setminus j}(t)\rangle_{i\setminus j} \tag{29a}$$

$$= \langle x_{i\setminus j}(t)\int dt''\left[2T\delta(t'-t'')\mathrm{i}\hat{x}_{i\setminus j}(t'') + \sum_{k\in\partial i\setminus j} J_{ik}^2 C_{k\setminus i}(t',t'')\mathrm{i}\hat{x}_{i\setminus j}(t'')\right]\rangle_{i\setminus j} \tag{29b}$$

$$= 2TR_{i\setminus j}(t,t') + \sum_{k\in\partial i\setminus j}\int_0^t dt''\, J_{ik}^2 C_{k\setminus i}(t',t'')R_{i\setminus j}(t,t''). \tag{29c}$$

Deriving the correlation function

$$\frac{\partial}{\partial t}C_{i\setminus j}(t,t') = \langle \dot{x}_{i\setminus j}(t)x_{i\setminus j}(t')\rangle_{i\setminus j} = C_{i\setminus j}(t,t') - \langle x_{i\setminus j}^2(t)x_{i\setminus j}(t')\rangle_{i\setminus j} + \sum_{k\in\partial i\setminus j} J_{ik}\mu_{k\setminus i}(t)C_{i\setminus j}(t,t')$$
$$+ \sum_{k\in\partial i\setminus j}\int_0^t dt''\, J_{ik}J_{ki}R_{k\setminus i}(t,t'')\langle x_{i\setminus j}(t)x_{i\setminus j}(t')x_{i\setminus j}(t'')\rangle_{i\setminus j} + \langle x_{i\setminus j}(t)x_{i\setminus j}(t')\xi_{i\setminus j}(t)\rangle_{i\setminus j} \tag{30a}$$

$$= C_{i\setminus j}(t,t') - \langle x_{i\setminus j}^2(t)x_{i\setminus j}(t')\rangle_{i\setminus j} + \sum_{k\in\partial i\setminus j} J_{ik}\mu_{k\setminus i}(t)C_{i\setminus j}(t,t') + 2T\langle x_{i\setminus j}(t)x_{i\setminus j}(t')\mathrm{i}\hat{x}_{i\setminus j}(t)\rangle_{i\setminus j}$$
$$+ \sum_{k\in\partial i\setminus j}\int_0^t dt''\, [J_{ik}J_{ki}R_{k\setminus i}(t,t'')\langle x_{i\setminus j}(t)x_{i\setminus j}(t')x_{i\setminus j}(t'')\rangle_{i\setminus j}$$
$$+ J_{ik}^2 C_{k\setminus i}(t,t'')\langle x_{i\setminus j}(t)x_{i\setminus j}(t')\mathrm{i}\hat{x}_{i\setminus j}(t'')\rangle_{i\setminus j}], \tag{30b}$$

and finally for the response

$$\frac{\partial}{\partial t}R_{i\setminus j}(t,t') = \langle \dot{x}_{i\setminus j}(t)\mathrm{i}\hat{x}_{i\setminus j}(t')\rangle_{i\setminus j} = R_{i\setminus j}(t,t') - \langle x_{i\setminus j}^2(t)\mathrm{i}\hat{x}_{i\setminus j}(t')\rangle_{i\setminus j} + \sum_{k\in\partial i\setminus j} J_{ik}\mu_{k\setminus i}(t)R_{i\setminus j}(t,t')$$
$$+ \sum_{k\in\partial i\setminus j}\int_0^t dt''\, J_{ik}J_{ki}R_{k\setminus i}(t,t'')\langle x_{i\setminus j}(t)\mathrm{i}\hat{x}_{i\setminus j}(t')x_{i\setminus j}(t'')\rangle_{i\setminus j} + \langle x_{i\setminus j}(t)\mathrm{i}\hat{x}_{i\setminus j}(t')\xi_{i\setminus j}(t)\rangle_{i\setminus j} \tag{31a}$$

$$= R_{i\setminus j}(t,t') - \langle x_{i\setminus j}^2(t)\mathrm{i}\hat{x}_{i\setminus j}(t')\rangle_{i\setminus j} + \sum_{k\in\partial i\setminus j} J_{ik}\mu_{k\setminus i}(t)R_{i\setminus j}(t,t') + 2T\langle x_{i\setminus j}(t)\mathrm{i}\hat{x}_{i\setminus j}(t')\mathrm{i}\hat{x}_{i\setminus j}(t)\rangle_{i\setminus j}$$
$$+ \sum_{k\in\partial i\setminus j}\int_0^t dt''\, [J_{ik}J_{ki}R_{k\setminus i}(t,t'')\langle x_{i\setminus j}(t)\mathrm{i}\hat{x}_{i\setminus j}(t')x_{i\setminus j}(t'')\rangle_{i\setminus j}$$
$$+ J_{ik}^2 C_{k\setminus i}(t,t'')\langle x_{i\setminus j}(t)\mathrm{i}\hat{x}_{i\setminus j}(t')\mathrm{i}\hat{x}_{i\setminus j}(t'')\rangle_{i\setminus j}], \tag{31b}$$

1. *Trying to approximate something*

Since we are interested in studying the statistics of the fixed-points of the system, we consider the long-time limit $t, t' \to \infty$. We can assume that the system retain no memory of the initial state distribution, therefore both the response and the correlation become Time Translational Invariant (TTI), i.e. $R(t,t') = R(\tau = t-t')$ and $C(t,t') = C(\tau = t-t')$. Eq. (28b) becomes

$$\frac{d}{dt}\mu_{i\setminus j}(t) = \mu_{i\setminus j}(t) - C_{i\setminus j}(0) + \sum_{k\in\partial i\setminus j} J_{ik}\mu_{k\setminus i}(t)\mu_{i\setminus j}(t)$$
$$+ \sum_{k\in\partial i\setminus j}\int d\tau\, \left[J_{ik}J_{ki}R_{k\setminus i}(\tau)C_{i\setminus j}(\tau) + J_{ik}^2 C_{k\setminus i}(\tau)R_{i\setminus j}(\tau)\right]. \tag{32}$$

Applying the Laplace transform to the expression above we obtain

$$
\begin{aligned}
z\hat{\mu}_{i\setminus j}(z) - \mu(0) = \hat{\mu}_{i\setminus j}(z) - C_{i\setminus j}(0)\delta(z) + \sum_{k\in\partial i\setminus j} J_{ik}\mathcal{LT}\left[\mu_{k\setminus i}(t)\mu_{i\setminus j}(t)\right] \\
+ \frac{1}{z}\sum_{k\in\partial i\setminus j}\left[J_{ik}J_{ki}\mathcal{LT}[R_{k\setminus i}(\tau)C_{k\setminus i}(\tau)] + J_{ik}^2\mathcal{LT}[C_{k\setminus i}(\tau)R_{i\setminus j}(\tau)]\right]. \qquad (33)
\end{aligned}
$$

Up to this point is essentially exact. Then, we decouple the Laplace transforms, assuming that $\mathcal{LT}[AB] \approx \mathcal{LT}[A]\mathcal{LT}[B]$. (**Within this approximation we are closing our eyes**)

$$
\begin{aligned}
z\hat{\mu}_{i\setminus j}(z) - \mu(0) = \hat{\mu}_{i\setminus j}(z) - C_{i\setminus j}(0)\delta(z) + \sum_{k\in\partial i\setminus j} J_{ik}\hat{\mu}_{k\setminus i}(z)\hat{\mu}_{i\setminus j}(z) \\
+ \frac{1}{z}\sum_{k\in\partial i\setminus j}\left[J_{ik}J_{ki}\hat{R}_{k\setminus i}(z)\hat{C}_{k\setminus i}(z)] + J_{ik}^2\hat{C}_{k\setminus i}(z)\hat{R}_{i\setminus j}(z)]\right]. \qquad (34)
\end{aligned}
$$

This leads to the self-consistent solution for μ

$$
\hat{\mu}_{i\setminus j}(z) = \frac{\mu(0) - C_{i\setminus j}(0)\delta(z) + \frac{1}{z}\sum_{k\in\partial i\setminus j}\left[J_{ik}J_{ki}\hat{R}_{k\setminus i}(z)\hat{C}_{k\setminus i}(z) + J_{ik}^2\hat{C}_{k\setminus i}(z)\hat{R}_{i\setminus j}(z)\right]}{z - 1 - \sum_{k\in\partial i\setminus j} J_{ik}\hat{\mu}_{k\setminus i}(z)}. \qquad (35)
$$

Notice that, for the Final Value Theorem (FVT), if $\mu_{i\setminus j}(t)$ is finite at infinite time, this finite value is defined by $\mu^{eq}_{i\setminus j} = \lim_{p\to 0} p\hat{\mu}_{i\setminus j}(p)$. Therefore

$$
\mu^{eq}_{i\setminus j} = -\frac{\sum_{k\in\partial i\setminus j}\left[J_{ik}J_{ki}\hat{R}_{k\setminus i}(0)\hat{C}_{k\setminus i}(0) + J_{ik}^2\hat{C}_{k\setminus i}(0)\hat{R}_{i\setminus j}(0)\right]}{\sum_{k\in\partial i\setminus j} J_{ik}\hat{\mu}_{k\setminus i}(0) + 1} \qquad (36)
$$

Here we have a big problem, if we assume that $\lim_{z\to 0} z\hat{\mu}(z) = \mu^{eq}$ exists and is finite, then $\hat{\mu}(z) \approx \mu^{eq}/z$ for $z \to 0$, which means that $\hat{\mu}(0)$ diverges ... It follows that the denominator diverges ...

We can do something similar for the remaining two quantities. Let us first focus on the response. First we must break down the correlation functions that involve more than two variables

$$
\begin{aligned}
\langle x^2_{i\setminus j}(t)\mathrm{i}\hat{x}_{i\setminus j}(t')\rangle_{i\setminus j} &\approx \langle x_{i\setminus j}(t)\rangle\langle x_{i\setminus j}(t)\mathrm{i}\hat{x}_{i\setminus j}(t')\rangle_{i\setminus j} = \mu_{i\setminus j}(t)R_{i\setminus j}(t,t') && (37a)\\
\langle x_{i\setminus j}(t)\mathrm{i}\hat{x}_{i\setminus j}(t')\mathrm{i}\hat{x}_{i\setminus j}(t)\rangle_{i\setminus j} &\approx \langle x_{i\setminus j}(t)\rangle\langle \mathrm{i}\hat{x}_{i\setminus j}(t')\mathrm{i}\hat{x}_{i\setminus j}(t)\rangle_{i\setminus j} = 0 && (37b)\\
\langle x_{i\setminus j}(t)\mathrm{i}\hat{x}_{i\setminus j}(t')x_{i\setminus j}(t'')\rangle_{i\setminus j} &\approx \left[\langle x_{i\setminus j}(t)\rangle\langle \mathrm{i}\hat{x}_{i\setminus j}(t')x_{i\setminus j}(t'')\rangle_{i\setminus j} + \langle x_{i\setminus j}(t'')\rangle\langle \mathrm{i}\hat{x}_{i\setminus j}(t)x_{i\setminus j}(t')\rangle_{i\setminus j}\right]/2 \\
&= [\mu_{i\setminus j}(t)R_{i\setminus j}(t'',t') + \mu_{i\setminus j}(t'')R_{i\setminus j}(t,t')]/2 && (37c)
\end{aligned}
$$

With this we can go further and rewrite the equation for R in terms of one and two point correlation functions alone.

$$
\begin{aligned}
\frac{\partial}{\partial t}R_{i\setminus j}(t,t') &= R_{i\setminus j}(t,t') - \mu_{i\setminus j}(t)R_{i\setminus j}(t,t') + \sum_{k\in\partial i\setminus j} J_{ik}\mu_{k\setminus i}(t)R_{i\setminus j}(t,t') \\
&\quad + \mu_{i\setminus j}(t)\sum_{k\in\partial i\setminus j}\int_0^t dt''\, J_{ik}J_{ki}R_{k\setminus i}(t,t'')R_{i\setminus j}(t',t'') \\
&= \left[1 - \mu_{i\setminus j}(t) + \sum_{k\in\partial i\setminus j} J_{ik}\mu_{k\setminus i}(t)\right]R_{i\setminus j}(t,t') + \mu_{i\setminus j}(t)\sum_{k\in\partial i\setminus j}\int_0^t dt''\, J_{ik}J_{ki}R_{k\setminus i}(t,t'')R_{i\setminus j}(t',t'') \qquad (38)
\end{aligned}
$$

For times large enough one can substitute above $\mu_{i\setminus j}(t) = \mu_{eq}$, and use again the time translational invariance to show that:

$$
\hat{R}_{i\setminus j}(p) = \frac{R_0}{p - \left[1 - \mu^{eq}_{i\setminus j} + \sum_{k\in\partial i\setminus j} J_{ik}\mu^{eq}_{k\setminus i}\right] - \mu^{eq}_{i\setminus j}\sum_{k\in\partial i\setminus j} J_{ik}J_{ki}R_{k\setminus i}(p)} \qquad (39)
$$

I feel that eqs (35), (36) and (39) are already similar in spirit to what you got before for the linear problem. **Please, check for a while the computations, and if they are not too crazy, we keep going**

ACKNOWLEDGMENTS

We wish to acknowledge the support of ....

---

* mattia.tarabolo@polito.it
[1] M. Opper and S. Diederich, Phys. Rev. Lett. **69**, 1616 (1992).
[2] T. Galla, J. Phys. A: Math. Gen. **39**, 3853 (2006).

Appendix A: Effective dynamics on a RRG

The calculation is based on the principles of [?]. The dynamical partition function of the Lotka-Volterra system can be written as

$$\mathrm{Z}{=}\left\langle \int \mathcal{D}\vec{x} \prod_i p_i\left(x_i\left(0\right)\right) \delta \left(\text{equation of motion}\right) \right\rangle_{\vec{\eta}}, \tag{A1}$$

where $\int \mathcal{D}\vec{x} = \prod_i \int \mathcal{D}x$ is a product of functional integrals, $\langle \dots \rangle_{\vec{\eta}} = \left(\prod_i \int \mathcal{D}\eta_i \right) \dots$ is the average over the noise realizations, and $\delta(\dots)$ is a functional Dirac delta function which restrict the integral to the paths that satisfy the stochastic dynamics. Using the Fourier representation of the delta function $\delta(x) \propto \int \mathcal{D}\hat{x}\, e^{-\mathrm{i} \int dt\, \hat{x}(t) x(t)}$ we obtain

$$Z = \left\langle \int \mathcal{D}\vec{x}\mathcal{D}\vec{\hat{x}} \prod_i p_i\left(x_i(0)\right) e^{\int dt\, (-\mathrm{i}\hat{x}_i(t)) \left[\dot{x_i}(t) - x_i(t)\left(1 - x_i - \sum_{j \neq i} a_{ij} J_{ij} x_j + h_i(t)\right) - \eta_i(t)\right]} \right\rangle_{\vec{\eta}}. \tag{A2}$$

The integral over time run over a fixed time interval, but we omit the limits of integration, here and elsewhere, for the sake of simplifying the notation. We introduce an auxiliary variable $f_i = 1 - x_i - \sum_{j \neq i} a_{ij} J_{ij} x_j(t) + h_i(t)$, which can be inserted in the calculation through a Dirac delta function. Integrating over the Gaussian noise we obtain

$$Z = \int \mathcal{D}\vec{x}\mathcal{D}\vec{\hat{x}}\mathcal{D}\vec{f}\mathcal{D}\vec{\hat{f}} \prod_{i \in V} p_i\left(x_i(0)\right) e^{S_i\left(x_i, \hat{x}_i, f_i, \hat{f}_i\right)} \prod_{(i,j) \in E} e^{S_{ij}\left(x_i, \hat{x}_i, f_i, \hat{f}_i, x_j, \hat{x}_j, f_j, \hat{f}_j\right)}, \tag{A3}$$

where we have defined the local and interactive actions as

$$S_i = \int dt \left\{ -\mathrm{i}\hat{x}_i(t)\left(\dot{x}_i(t) - x_i(t) f_i(t)\right) - \frac{1}{2} g(x_i(t)) \hat{x}_i^2(t) - \mathrm{i}\hat{f}_i(t)\left[f_i(t) - \left(1 - x_i(t) + h_i(t)\right)\right] \right\} \tag{A4}$$

$$S_{ij} = -\int dt \left(J_{ij} \mathrm{i}\hat{f}_i(t) x_j(t) + J_{ji} \mathrm{i}\hat{f}_j(t) x_i(t) \right) \tag{A5}$$

We put forward the following *dynamic cavity equations* as an ansatz for describing linearly coupled stochastic dynamics on sparse networks

$$c_{i\setminus j}[x_i, \hat{x}_i, f_i, \hat{f}_i] = \frac{1}{Z_{i\setminus j}} e^{S_i} \prod_{k \in \partial i \setminus j} \int \mathcal{D}x_k \mathcal{D}\hat{x}_k \mathcal{D}f_k \mathcal{D}\hat{f}_k\, c_{k\setminus i}[x_k, \hat{x}_k, f_k, \hat{f}_k] e^{S_{ki}}, \tag{A6}$$

The full dynamic cavity marginals can be obtained from the cavity ones as follows

$$c_i[x_i, \hat{x}_i, f_i, \hat{f}_i] = \frac{1}{Z_i} e^{S_i} \prod_{k \in \partial i} \int \mathcal{D}x_k \mathcal{D}\hat{x}_k \mathcal{D}f_k \mathcal{D}\hat{f}_k\, c_{k\setminus i}[x_k, \hat{x}_k, f_k, \hat{f}_k] e^{S_{ki}}. \tag{A7}$$

Expanding the cavity marginals Eq. (A6) at second order in the coupling parameters J_{ij} we obtain an effective single node action for the cavity marginal

$$c_{i\setminus j}[x_i,\hat{x}_i,f_i,\hat{f}_i]\approx\frac{p_i(x_i(0))}{Z_{i\setminus j}}e^{S^{\mathrm{eff}}_{i\setminus j}},\tag{A8}$$

where the effective cavity action is defined as

$$\begin{aligned}S^{\mathrm{eff}}_{i\setminus j}=\int dt\Bigg\{&-\mathrm{i}\hat{x}_i(t)\left(\dot{x}_i(t)-x_i(t)f_i(t)\right)-\frac{1}{2}g(x_i(t))\hat{x}_i^2(t)-\mathrm{i}\hat{f}_i(t)\Big[f_i(t)-(1-x_i(t)+h_i(t))\\&+\sum_{k\in\partial i\setminus j}J_{ik}\mu_{k\setminus i}(t)-\sum_{k\in\partial i\setminus j}J_{ik}J_{ki}\int dt'\,R_{k\setminus i}(t,t')x_i(t')\Big]\Bigg\}+\int dtdt'\,\mathrm{i}\hat{f}(t)\left(\sum_{k\in\partial i\setminus j}J^2_{ik}C_{k\setminus i}(t,t')\right)\mathrm{i}\hat{f}(t')\end{aligned}\tag{A9}$$

where $\mu_{k\setminus i}(t)$, $R_{k\setminus i}(t,t')$, and $C_{k\setminus i}(t,t')$ are respectively the cavity mean abundance, the cavity response function, and the cavity (connected) two-point correlation function. They are obtained self-consistently from the effective dynamics of the neighbors

$$\mu_{k\setminus i}(t)=\langle x_k(t)\rangle_{k\setminus i}\,,\quad R_{k\setminus i}(t,t')=\left\langle x_k(t)\mathrm{i}\hat{f}_k(t')\right\rangle_{k\setminus i}\,,\quad C(t,t')=\langle x_k(t)x_k(t')\rangle_{k\setminus i}-\mu_{k\setminus i}(t)\mu_{k\setminus i}(t'),\tag{A10}$$

where $\langle\ldots\rangle_{k\setminus i}=\int\mathcal{D}x_k\mathcal{D}\hat{x}_k\mathcal{D}f_k\mathcal{D}\hat{f}_k\,c_{k\setminus i}[x_k,\hat{x}_k,f_k,\hat{f}_k]\ldots$ describes the local average over the cavity marginal. The effective cavity action can be identified with a sort of local effective single-edge action for the variables on site i when the interaction term with site j is removed from the underlying graph.

Performing backward the steps of the path-integral representation, we obtain an effective stochastic dynamics in terms of the stochastic differential equation on the cavity graph

$$\begin{aligned}\frac{d}{dt}x_{i\setminus j}(t)=x_{i\setminus j}(t)\Bigg[&1-x_{i\setminus j}(t)+\sum_{k\in\partial i\setminus j}J_{ik}\mu_{k\setminus i}(t)+\sum_{k\in\partial i\setminus j}\int_0^t dt'\,J_{ik}J_{ki}R_{k\setminus i}(t,t')x_{i\setminus j}(t')\\&+h_{i\setminus j}(t)+\zeta_{i\setminus j}(t)\Bigg]+\xi_{i\setminus j}(t),\end{aligned}\tag{A11}$$

where $\zeta_{i\setminus j}$ and $\xi_{i\setminus j}$ are Gaussian noises with vanishing mean and correlations

$$\left\langle\zeta_{i\setminus j}(t)\zeta_{i\setminus j}(t')\right\rangle_{i\setminus j}=\sum_{k\in\partial i\setminus j}J^2_{ik}C_{k\setminus i}(t,t'),\qquad\left\langle\xi_{i\setminus j}(t)\xi_{i\setminus j}(t')\right\rangle_{i\setminus j}=g(x_{i\setminus j}(t))\delta(t-t').\tag{A12}$$

The terms $\mu_{k\setminus i}(t)$, $R_{k\setminus i}(t,t')$, and $C_{k\setminus i}(t,t')$ have to be considered as external parameters, obtained self-consistently as averages over the effective dynamics of the neighbors. In a similar way, an effective process for the original graph can be obtained as

$$\begin{aligned}\frac{d}{dt}x_i(t)=x_i(t)\Bigg[&1-x_i(t)+\sum_{k\in\partial i}J_{ik}\mu_{k\setminus i}(t)+\sum_{k\in\partial i}\int_0^t dt'\,J_{ik}J_{ki}R_{k\setminus i}(t,t')x_i(t')\\&+h_i(t)+\zeta_i(t)\Bigg]+\xi_i(t),\end{aligned}\tag{A13}$$

with Gaussian noises with vanishing mean and correlations

$$\langle\zeta_i(t)\zeta_i(t')\rangle_i=\sum_{k\in\partial i}J^2_{ik}C_{k\setminus i}(t,t'),\qquad\langle\xi_i(t)\xi_i(t')\rangle_i=g(x_i(t))\delta(t-t').\tag{A14}$$

Appendix B: Gaussian expansion of the disorder averaged dynamical partition function

We focus here on the case of an heterogeneous random graph G, where edges are randomly extracted from an adjacency matrix distribution $p(A)$. A single instance G of the random graph ensemble has a set of fixed degrees for each node $k_i = \sum_j a_{ij}$, which are sampled from a particular degree distribution $p(k)$. We denote the average degree of the ensemble as $K = \sum_k kp(k)$. We can write the distribution of the adjacency matrix as

$$p(a_{ij}, a_{ji}) = \frac{K}{N}\delta_{a_{ij},1}\delta_{a_{ji},1} + \left(1 - \frac{K}{N}\right)\delta_{a_{ij},0}\delta_{a_{ji},0}.$$

We use the Gaussian Expansion Cavity method [?] to derive a Langevin equation for a representative species. This approach begins by constructing the Martin-Siggia-Rose-Janssen-De Dominicis (MSRJD) dynamic partition function through path integrals of the trajectories [? ? ?]

$$Z = \overline{\left\langle \int \mathcal{D}\vec{x} \prod_i \delta\left(\frac{\dot{x}_i(t)}{x_i(t)} - \left[1 - x_i(t) + \sum_{j\neq i} a_{ij}J_{ij}x_j(t) + \eta_i(t) + h_i(t)\right]\right)\delta\left(\sum_j a_{ij} - k_i\right)\right\rangle}_{\vec{\eta}} \tag{B1a}$$

$$\propto \overline{\left\langle \int \mathcal{D}\vec{x}\mathcal{D}\vec{\hat{x}}d\vec{\psi}\prod_i e^{\int dt\, \mathrm{i}\hat{x}_i\left(\frac{\dot{x}_i}{x_i} - \left[1-x_i+\sum_{j\neq i}a_{ij}J_{ij}x_j+\eta_i+h_i\right]\right)} e^{\mathrm{i}\psi_i\left(\sum_j a_{ij}-k_i\right)}\right\rangle}_{\vec{\eta}} \tag{B1b}$$

$$\propto \int \mathcal{D}\vec{x}\mathcal{D}\vec{\hat{x}}d\vec{\psi}\prod_i e^{\int dt\left[\mathrm{i}\hat{x}_i\left(\frac{\dot{x}_i}{x_i}-1+x_i+h_i\right)-T\hat{x}_i^2\right]} e^{-\mathrm{i}\psi_i k_i}\overline{\prod_{i<j} e^{[a_{ij}(\mathrm{i}\psi_i-\int dt\, J_{ij}\mathrm{i}\hat{x}_i x_j)+a_{ji}(\mathrm{i}\psi_j-\int dt\, J_{ji}\mathrm{i}\hat{x}_j x_i)]}} \tag{B1c}$$

$$= \int \mathcal{D}\vec{x}\mathcal{D}\vec{\hat{x}}d\vec{\psi}\prod_i e^{\int dt\left[\mathrm{i}\hat{x}_i\left(\frac{\dot{x}_i}{x_i}-1+x_i+h_i\right)-T\hat{x}_i^2\right]} e^{-\mathrm{i}\psi_i k_i}\prod_{i<j}\left[1-\frac{K}{N}+\frac{K}{N}\overline{e^{[(\mathrm{i}\psi_i-\int dt\, J\mathrm{i}\hat{x}_i x_j)+(\mathrm{i}\psi_j-\int dt\, J'\mathrm{i}\hat{x}_j x_i)]}}\right] \tag{B1d}$$

$$\approx \int \mathcal{D}\vec{x}\mathcal{D}\vec{\hat{x}}d\vec{\psi}\prod_i e^{\int dt\left[\mathrm{i}\hat{x}_i\left(\frac{\dot{x}_i}{x_i}-1+x_i+h_i\right)-T\hat{x}_i^2\right]} e^{-\mathrm{i}\psi_i k_i} e^{-\frac{KN}{2}+\frac{K}{2N}\sum_{ij}\overline{e^{-\int dt\left(J\mathrm{i}\hat{x}_i x_j+J'\mathrm{i}\hat{x}_j x_i\right)}}e^{\mathrm{i}(\psi_i+\psi_j)}}. \tag{B1e}$$

We introduce the usual functional order parameter following the steps of [?]

$$c[x,\hat{x}] = \frac{1}{N}\sum_i e^{-\mathrm{i}\psi_i}\delta(x-x_i)\delta(\hat{x}-\hat{x}_i). \tag{B2}$$

This definition can be inserted into Z using as usual functional Dirac deltas

$$Z \propto \int \mathcal{D}\vec{x}\mathcal{D}\vec{\hat{x}}d\vec{\psi}\prod_i e^{\int dt\left[\mathrm{i}\hat{x}_i\left(\frac{\dot{x}_i}{x_i}-1+x_i+h_i\right)-T\hat{x}_i^2\right]} e^{-\mathrm{i}\psi_i k_i} e^{\frac{K}{2N}\sum_{ij}\overline{e^{-\int dt\left(J\mathrm{i}\hat{x}_i x_j+J'\mathrm{i}\hat{x}_j x_i\right)}}e^{\mathrm{i}(\psi_i+\psi_j)}}$$
$$\times \int \mathcal{D}c\,\delta\left(Nc[x,\hat{x}] - \sum_i e^{-\mathrm{i}\psi_i}\delta(x-x_i)\delta(\hat{x}-\hat{x}_i)\right) \tag{B3a}$$

$$= \int \mathcal{D}\vec{x}\mathcal{D}\vec{\hat{x}}d\vec{\psi}\prod_i e^{\int dt\left[\mathrm{i}\hat{x}_i\left(\frac{\dot{x}_i}{x_i}-1+x_i+h_i\right)-T\hat{x}_i^2\right]} e^{-\mathrm{i}\psi_i k_i} e^{\frac{K}{2N}\sum_{ij}\overline{e^{-\int dt\left(J\mathrm{i}\hat{x}_i x_j+J'\mathrm{i}\hat{x}_j x_i\right)}}e^{\mathrm{i}(\psi_i+\psi_j)}}$$
$$\times \int \mathcal{D}c\mathcal{D}\hat{c}\, e^{\mathrm{i}\int \mathcal{D}x\mathcal{D}\hat{x}\,\hat{c}[x,\hat{x}]\left(Nc[x,\hat{x}]-\sum_i e^{-\mathrm{i}\psi_i}\delta(x-x_i)\delta(\hat{x}-\hat{x}_i)\right)} \tag{B3b}$$

$$= \int \mathcal{D}c\mathcal{D}\hat{c}\, e^{\mathrm{i}N\int \mathcal{D}x\mathcal{D}\hat{x}\,\hat{c}[x,\hat{x}]c[x,\hat{x}]+\frac{KN}{2}\int \mathcal{D}x\mathcal{D}\hat{x}\,c[x,\hat{x}]\int \mathcal{D}y\mathcal{D}\hat{y}\,c[y,\hat{y}]\overline{e^{-\int dt\left(\mathrm{i}\hat{x}Jy+\mathrm{i}\hat{y}J'x\right)}}}$$
$$\times \prod_k\left[\int \mathcal{D}x\mathcal{D}\hat{x}e^{\int dt\left[\mathrm{i}\hat{x}\left(\frac{\dot{x}}{x}-1+x+h\right)-T\hat{x}^2\right]}\frac{\left(-\mathrm{i}\hat{c}\,[x,\hat{x}]\right)^k}{k!}\right]^{N_k}. \tag{B3c}$$

We end up with the usual heterogeneous dynamical mean field theory, already done by [?], .

$$Z \propto \int \mathcal{D}c\mathcal{D}\hat{c}\, e^{N\Psi[c,\hat{c}]} \sim e^{N\Psi^*}, \tag{B4}$$

where

$$\Psi^* = \max_{c,\hat{c}} \left\{ \Xi_1[c,\hat{c}] + \Xi_2[c] + \Xi_3[\hat{c}] \right\}, \tag{B5}$$

and

$$\Xi_1[c,\hat{c}] = \mathrm{i} \int \mathcal{D}x\mathcal{D}\hat{x}\, \hat{c}[x,\hat{x}]c[x,\hat{x}] \tag{B6a}$$

$$\Xi_2[c] = \frac{K}{2} \int \mathcal{D}x\mathcal{D}\hat{x}\, c[x,\hat{x}] \int \mathcal{D}y\mathcal{D}\hat{y}\, c[y,\hat{y}]\overline{e^{-\int dt\,(\mathrm{i}\hat{x}Jy + \mathrm{i}\hat{y}J'x)}} \tag{B6b}$$

$$\begin{aligned} \Xi_3[\hat{c}] &= \frac{1}{N} \log \prod_k \left[\int \mathcal{D}x\mathcal{D}\hat{x} e^{\int dt\,\left[\mathrm{i}\hat{x}\left(\frac{\dot{x}}{x} - 1 + x + h\right) - T\hat{x}^2\right]} \frac{\left(-\mathrm{i}\hat{c}\,[x,\hat{x}]\right)^k}{k!} \right]^{N_k} \\ &\sim \sum_k p(k) \log \int \mathcal{D}x\mathcal{D}\hat{x} e^{\int dt\,\left[\mathrm{i}\hat{x}\left(\frac{\dot{x}}{x} - 1 + x + h\right) - T\hat{x}^2\right]} \frac{\left(-\mathrm{i}\hat{c}\,[x,\hat{x}]\right)^k}{k!}. \end{aligned} \tag{B6c}$$

The saddle-point solution gives

$$\hat{c}[x,\hat{x}] = \mathrm{i}K \int \mathcal{D}y\mathcal{D}\hat{y}\, c[y,\hat{y}]\overline{e^{-\int dt\,(\mathrm{i}\hat{x}Jy + \mathrm{i}\hat{y}J'x)}} \tag{B7a}$$

$$c[x,\hat{x}] = K \sum_k \frac{kp(k)}{K} \frac{e^{\int dt\,\left[\mathrm{i}\hat{x}\left(\frac{\dot{x}}{x} - 1 + x + h\right) - T\hat{x}^2\right]} \left(-\mathrm{i}\hat{c}\,[x,\hat{x}]\right)^{k-1}}{\int \mathcal{D}x\mathcal{D}\hat{x}\, e^{\int dt\,\left[\mathrm{i}\hat{x}\left(\frac{\dot{x}}{x} - 1 + x + h\right) - T\hat{x}^2\right]} \left(-\mathrm{i}\hat{c}\,[x,\hat{x}]\right)^k}. \tag{B7b}$$

Eliminating $\hat{c}$ leads to the following self-consistent equation for the functional order parameter c

$$c[x,\hat{x}] = \sum_k \frac{kp(k)}{K} \frac{1}{Z_{\mathrm{eff},k}} e^{-S_{\mathrm{eff},k-1}[x,\hat{x}]}, \tag{B8}$$

with the effective local action

$$\begin{aligned} S_{\mathrm{eff},k}[x,\hat{x}] = &- \int dt\, \left[\mathrm{i}\hat{x} \left(\frac{\dot{x}}{x} - 1 + x + h \right) - T\hat{x}^2 \right] \\ &- k \log \int \mathcal{D}y\mathcal{D}\hat{y}\, c[y,\hat{y}]\overline{e^{-\int dt\,(\mathrm{i}\hat{x}Jy + \mathrm{i}\hat{y}J'x)}} \end{aligned} \tag{B9a}$$

$$Z_{\mathrm{eff},k} = \int \mathcal{D}x\mathcal{D}\hat{x}\, e^{-S_{\mathrm{eff},k}[x,\hat{x}]} = 1. \tag{B9b}$$

The partition function ensures the desired normalization of the functional order parameter

$$\int \mathcal{D}x\mathcal{D}\hat{x}\, c[x,\hat{x}] = 1. \tag{B10}$$

Expanding up to second order in J the effective local action

$$\begin{aligned} S_{\mathrm{eff},k}[x,\hat{x}] \approx &- \int dt\, \left[\mathrm{i}\hat{x} \left(\frac{\dot{x}}{x} - 1 + x + h \right) - T\hat{x}^2 \right] + \mathrm{i}k\overline{J} \int dt\, \left(\hat{x}\langle y \rangle + \langle \hat{y} \rangle x \right) \\ &- \frac{k}{2} \int dt dt'\, \left(\overline{J^2}\hat{x}(t)\langle y(t)y(t')\rangle\hat{x}(t') + 2\overline{JJ'}\hat{x}(t)\langle y(t)\hat{y}(t')\rangle x(t') + \overline{J'^2}x(t)\langle \hat{y}(t)\hat{y}(t')\rangle x(t') \right) \end{aligned} \tag{B11a}$$

$$\begin{aligned} = &- \int dt\, \left[\mathrm{i}\hat{x} \left(\frac{\dot{x}}{x} - 1 + x + h \right) - T\hat{x}^2 \right] + \mathrm{i}k\overline{J} \int dt\, \hat{x}(t)M_{\mathrm{cav}}(t) \\ &- \frac{k}{2} \int dt dt'\, \left(\overline{J^2}\hat{x}(t)C_{\mathrm{cav}}(t,t')\hat{x}(t') + 2\overline{JJ'}\hat{x}(t)R_{\mathrm{cav}}(t,t')x(t') \right) \end{aligned} \tag{B11b}$$

Appendix C: Fixed point derivation on a RRG

The cavity fixed point solution is given by (24). Since ξ_c^* is Gaussian with zero average and variance $\langle(\xi_c^*)^2\rangle = \frac{K-1}{K}\left(\frac{m^2}{K}+\sigma^2\right)q_c^* = \Sigma_c^2 q_c^*$ we can write $\xi_c^* = -\Sigma_c\sqrt{q_c^*}s$ where s is a standard Gaussian random variable with zero mean and unit variance. Defining $\Delta_c = \left(1-\frac{K-1}{K}m\mu_c^*\right)/\left(\Sigma_c\sqrt{q_c^*}\right)$ and assuming $1-\frac{K-1}{K}(\frac{m^2}{K}+\gamma\sigma^2)\chi_c^* > 0$ we can write

$$x_c^*(s) = \frac{\Sigma_c\sqrt{q_c^*}\,(\Delta_c - s) + h_c}{1-\frac{K-1}{K}(\frac{m^2}{K}+\gamma\sigma^2)\chi_c^*}\Theta\left(\Sigma_c\sqrt{q_c^*}\,(\Delta_c - s) + h_c\right). \tag{C1}$$

Averaging over the distribution of $s$at $h_c = 0$

$$\mu_c^* = \langle x_c^*(s)\rangle_s = \frac{\Sigma_c\sqrt{q_c^*}}{1-\frac{K-1}{K}\left(\frac{m^2}{K}+\gamma\sigma^2\right)\chi_c^*}\int_{-\infty}^{\Delta_c} Ds\ (\Delta_c - s) \tag{C2a}$$

$$q_c^* = \langle (x_c^*(s))^2\rangle_s = \frac{\Sigma_c^2 q_c^*}{\left[1-\frac{K-1}{K}\left(\frac{m^2}{K}+\gamma\sigma^2\right)\chi_c^*\right]^2}\int_{-\infty}^{\Delta_c} Ds\ (\Delta_c - s)^2\,, \tag{C2b}$$

where $Ds = e^{-s^2/2}/\sqrt{2\pi}$. The integrated response χ_c^* is obtained as derivative of the average cavity population size with respect to a constant vanishing field, acting at all times, i.e. $h_c(t) = h_c$

$$\chi_c^* = \frac{d}{dh_c}\langle x_c^*(s)\rangle_s\bigg|_{h_c=0} = \frac{d}{dh_c}\frac{1}{1-\frac{K-1}{K}\left(\frac{m^2}{K}+\gamma\sigma^2\right)\chi_c^*}\int_{-\infty}^{\Delta_c + h_c/(\Sigma_c\sqrt{q_c^*})} Ds\ \left[\Sigma_c\sqrt{q_c^*}\,(\Delta_c - s) + h_c\right] \tag{C3}$$

$$= \frac{1}{1-\frac{K-1}{K}\left(\frac{m^2}{K}+\gamma\sigma^2\right)\chi_c^*}\int_{-\infty}^{\Delta_c} Ds. \tag{C4}$$

Appendix D: Stability analysis on a RRG

In order to study the stability of the fixed points, we perform a linear stability analysis. We add white noise $\zeta(t)$ of unit variance to the cavity effective process

$$\frac{d}{dt}x_c(t) = x_c(t)\Bigg[1 - x_c(t) + \frac{K-1}{K}m\mu_c(t) + \frac{K-1}{K}\left(\frac{m^2}{K}+\gamma\sigma^2\right)\int_0^t dt' R_c(t,t')x_c(t') + \xi_c(t) + h_c(t) + \varepsilon\zeta(t)\Bigg]. \tag{D1}$$

We study fluctuations $y(t)$ about a fixed point of Eq. (9), i.e. we write $x_c(t) = x_c^* + \varepsilon y(t)$, and denote the resulting additional term in the self-consistent noise by $\varepsilon\nu(t)$ (i.e., $\xi_c(t) = \xi_c^* + \varepsilon\nu(t)$). We use $\langle\zeta(t)\rangle = \langle\nu(t)\rangle = \langle y(t)\rangle = 0$.

a. Extinct species

We consider firstly the stability of the fixed point $x_c^* = 0$. Linearizing Eq. (D1) in ε

$$\frac{d}{dt}y(t) = y(t)\left(1 + \frac{K-1}{K}m\mu_c^* + \xi_c^*\right). \tag{D2}$$

Given our assumptions, the expression within the square brackets is negative for fixed points at zero [refer to Eq. (24)]. This implies that any disturbances around these zero fixed points diminish exponentially over time. Conversely, if the expression in the square brackets is positive, the zero fixed point becomes unstable, validating our choice to instead use the non-zero solution x_c^* in such cases.

b. *Extant species*

We now focus on extant species, i.e. non-vanishing fixed points $x_c^* > 0$. Linearizing Eq. (D1) in ε

$$\frac{d}{dt}y(t) = x_c^* \left[-y(t) + \frac{K-1}{K}\left(\frac{m^2}{K} + \gamma\sigma^2\right)\int_0^t dt'\, R_c^*(t,t')y(t') + \nu(t) + \zeta(t) \right], \tag{D3}$$

where $\langle \nu(t)\nu(t')\rangle = \frac{K-1}{K}(\frac{m^2}{K} + \sigma^2)\langle y(t)y(t')\rangle$ by self-consistency. Passing to Fourier space we obtain

$$\frac{\mathrm{i}\omega}{x_c^*}\tilde{y}(\omega) = \left[\frac{K-1}{K}(\frac{m^2}{K} + \gamma\sigma^2)\tilde{R}_c^*(\omega) - 1\right]\tilde{y}(\omega) + \tilde{\nu}(\omega) + \tilde{\zeta}(\omega), \tag{D4}$$

from which we can write

$$\tilde{y}(\omega) = \frac{\tilde{\nu}(\omega) + \tilde{\zeta}(\omega)}{\frac{\mathrm{i}\omega}{x_c^*} + \left(1 - \frac{K-1}{K}\left(\frac{m^2}{K} + \gamma\sigma^2\right)\tilde{R}_c^*(\omega)\right)}.$$

Noting that $\langle|\tilde{y}(\omega)|^2\rangle$ is the Fourier transform of the correlation of the fluctuations $\langle y(t)y(t+\tau)\rangle$ in the stationary state where it only depends on τ, we can study its long-time behaviour at $\omega = 0$. If this quantity diverges, perturbations do not decay to zero, signalling an instability of the fixed point. Using $\tilde{R}_c^*(\omega) = \int d\tau\, R_c^*(\tau) = \chi_c^*$

$$\langle|\tilde{y}(0)|^2\rangle = \frac{\phi_c}{\left[1 - \frac{K-1}{K}\left(\frac{m^2}{K} + \gamma\sigma^2\right)\chi_c^*\right]^2 - \frac{K-1}{K}\left(\frac{m^2}{K} + \sigma^2\right)\phi_c}, \tag{D5}$$

where $\phi_c = \int_{-\infty}^{\Delta_c} Ds$, fraction of surviving species in the cavity process, appears because we are only considering non-zero fixed points (fluctuations about zero fixed points decay, as demonstrated above, and so they do not contribute to $\langle|\tilde{y}(0)|^2\rangle$.

The correlation diverges when $\frac{K-1}{K}\left(\frac{m^2}{K} + \sigma^2\right)\phi_c = \left[1 - \frac{K-1}{K}\left(\frac{m^2}{K} + \gamma\sigma^2\right)\chi_c^*\right]^2$. Inserting the critical condition into Eq. (15b) we find

$$\phi_c = \int_{-\infty}^{\Delta_c} Ds\,(\Delta_c - s)^2 = \int_{-\infty}^{\Delta_c} Ds, \tag{D6}$$

from which $\Delta_c = 0$ and $\phi_c = 1/2$. Inserting these values into Eqs. (15b) and (15c) and eliminating χ_c^* we obtain the critical condition

$$\frac{\left(2\frac{m^2}{K} + (1+\gamma)\sigma^2\right)^2}{\frac{m^2}{K} + \sigma^2} = 2\frac{K}{K-1}. \tag{D7}$$